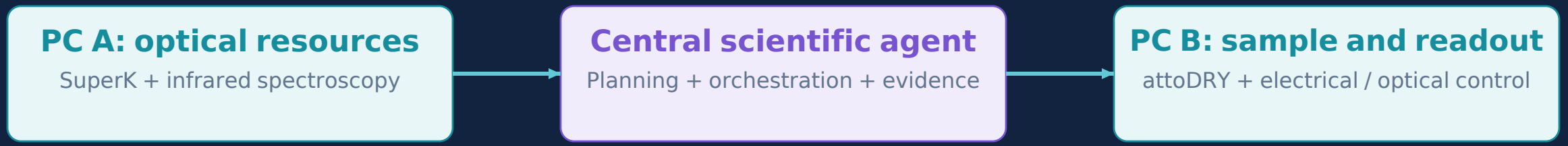
DISTRIBUTED EXPERIMENTS

# LabOS

## One experiment. Multiple PCs.
## A connected measurement system.

Nonlinear Hall transport + bulk photovoltaic response




A system-level view of hardware, references, timing, data and scientific decisions

# Two response channels, one experimental context

Electrical and optical excitation share the same sample coordinates: T, B and gate voltage Vg.

## Nonlinear Hall transport

Drive: lock-in sine output + series resistor
Readout: longitudinal Vxx and transverse Vxy
Axes: current, frequency, T, B and Vg
Controls: harmonics, return points, repeatability

## BVPE / photocurrent response

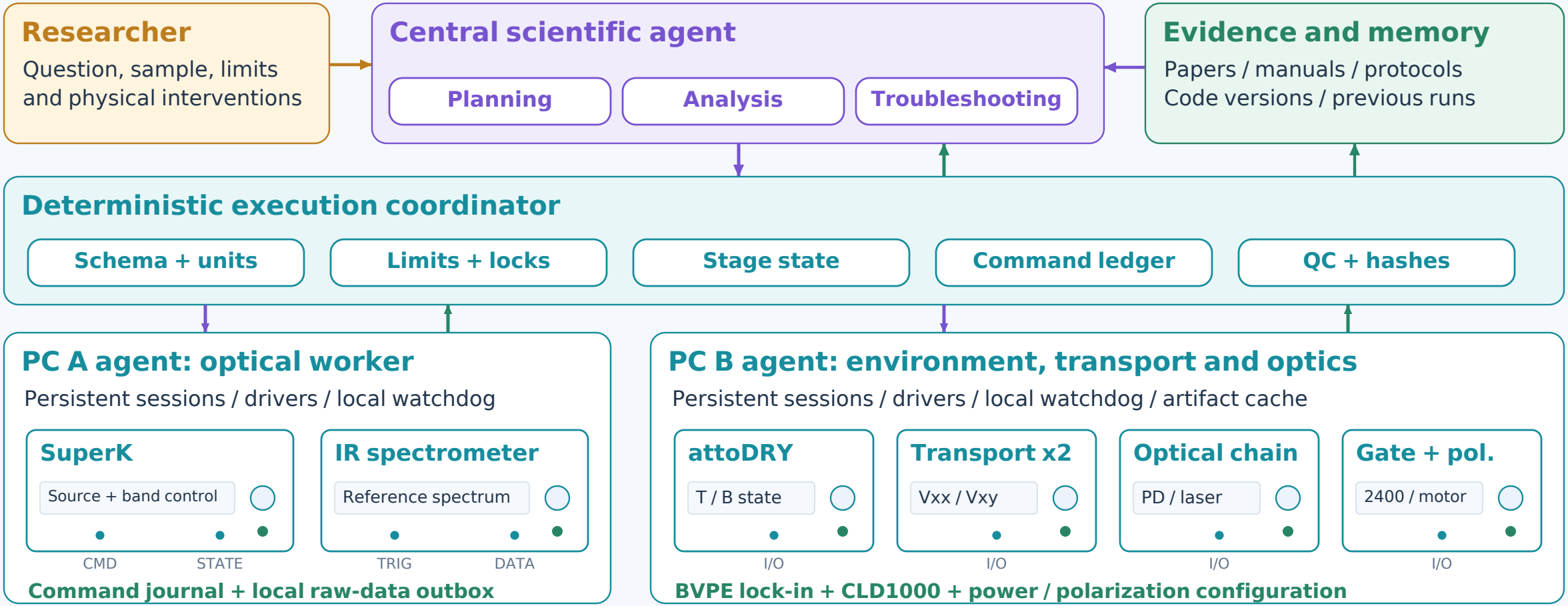
Drive: SuperK or a 730 nm laser
Reference: optical chopper at fchop
Axes: optical power, polarization, spectrum, T/B/Vg
Controls: dark, power reference and repeat scans

**Shared sample** | **Readback coordinates** | **Controlled stages** | **Linked evidence**

**The research task defines the sequence; device ownership defines where each action executes.**

# An evidence-driven hierarchy across two control PCs

Scientific reasoning, deterministic orchestration and local instrument ownership are separate layers.

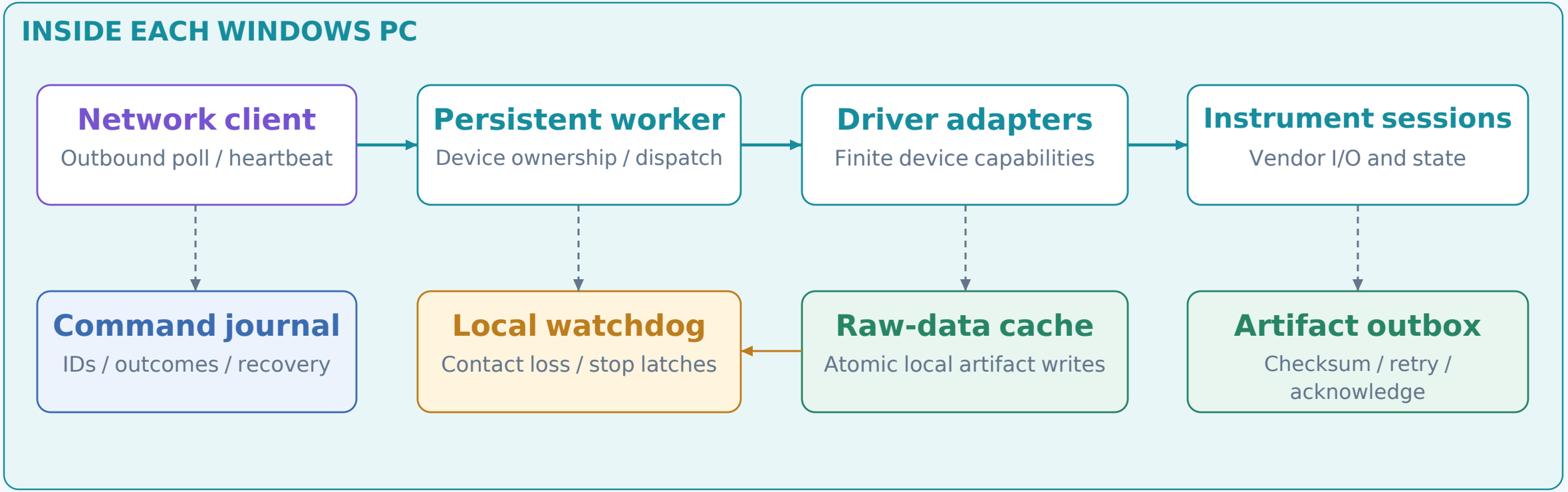



DOWN: approved commands  UP: data, readbacks and faults  LOCAL: session ownership and recovery state

Architecture presentation inspired by QUANTA Fig. 1 [1]; original LabOS redraw for this instrument scenario.

# A PC agent is a persistent instrument service

The central server schedules stages; the worker owns sessions, verifies actions and protects local evidence.




**A network interruption does not erase a command, transfer ownership or discard a saved measurement.**

# The complete two-PC instrument inventory

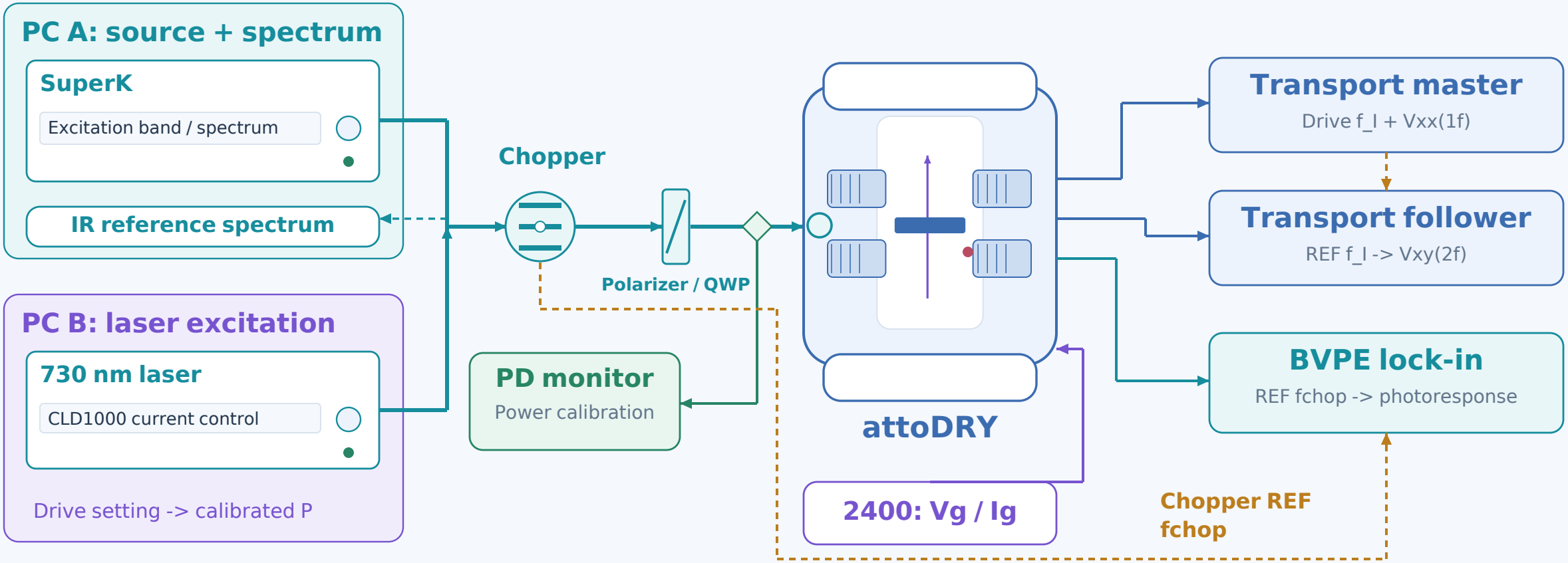
Device ownership stays local; instrument capabilities can participate in a shared multi-PC experiment.

| Owner | Instrument / resource | Experiment role |
|---|---|---|
| PC A | SuperK supercontinuum source | Optical source and excitation-band configuration |
| PC A | Infrared spectrometer | Incident / reference spectrum and optical diagnostics |
| PC B | attoDRY | Temperature, magnetic field and sample-state readbacks |
| PC B | Transport lock-in x2 | Master drive + Vxx; Vxy with selected harmonic |
| PC B | Chopper + BVPE lock-in | Optical modulation and referenced photoresponse |
| PC B | Photodiode (PD) | Optical-power monitoring and sample-plane calibration |
| PC B | 2400 source-measure unit | Gate voltage Vg and gate leakage Ig |
| PC B | 730 nm laser + CLD1000 | Laser excitation and drive-current control |
| PC B | Motorized polarizer / quarter-wave plate | Linear-polarization angle / polarization state |

**Two transport lock-ins + one BVPE lock-in; one configured source path and one shared sample context.**

# One sample connects the optical and electrical chains

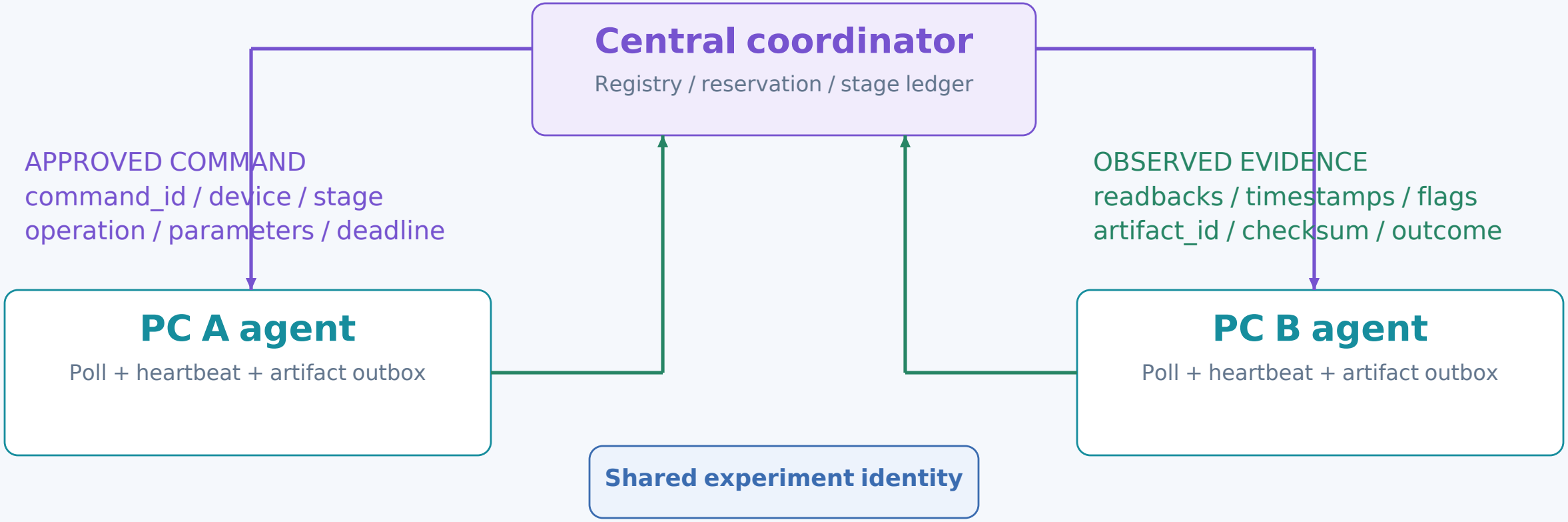
Control ownership follows PCs; light, voltage, current and physical references follow the apparatus.




**Legend: teal = optical / photoresponse; blue = transport readout; amber dashed = physical reference.**

# Orchestration travels over HTTP(S), not signal cables

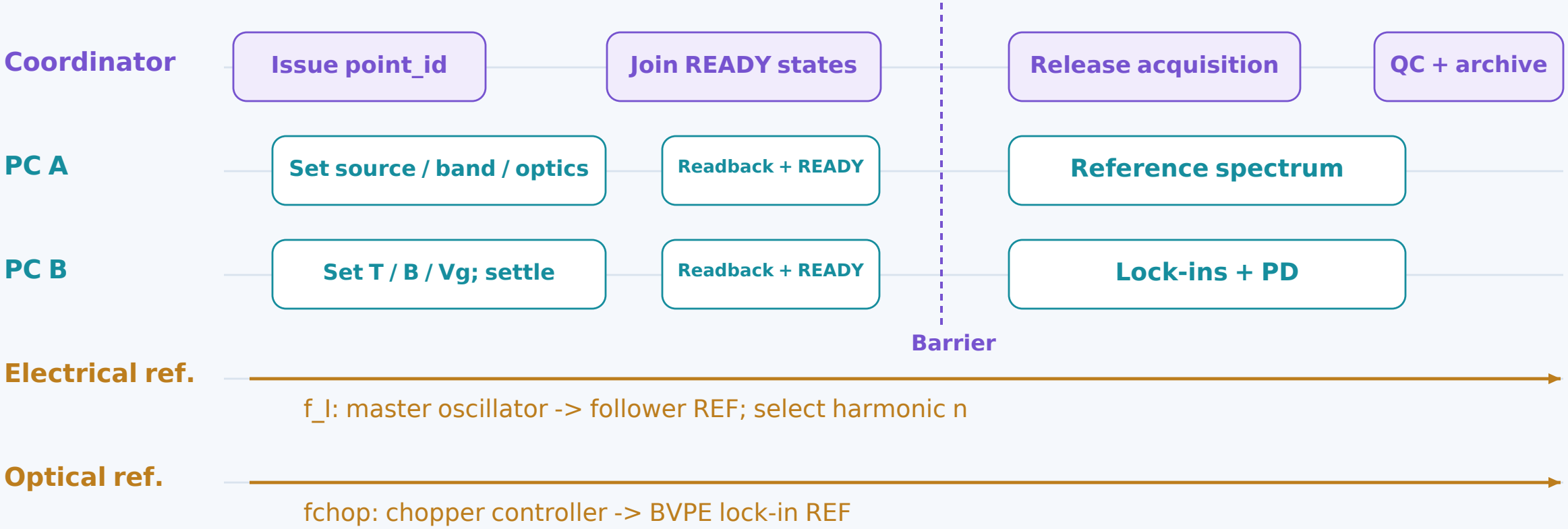
Windows agents initiate outbound connections; device commands are durable, scoped and attributable.




**No SSH dependency: the network coordinates state; physical references establish lock-in phase.**

# A point barrier links conditions before acquisition

A shared point_id provides correspondence; acquisition intervals and physical references provide timing context.




**Store conditions AND time windows. Shared coordination is not a claim of zero-skew sampling.**

# Translate the apparatus into a reproducible run setup

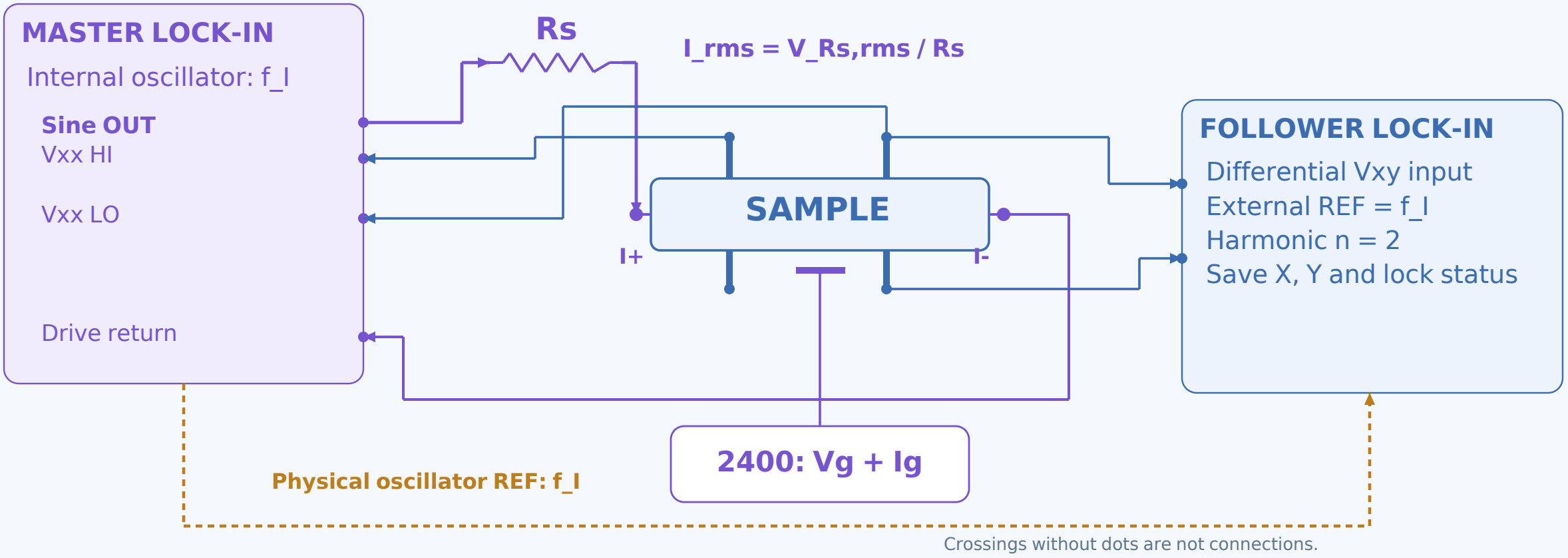
Reusable device capabilities are combined with sample-specific wiring, calibration and experimental limits.

| Record | Configuration content | Used by |
|---|---|---|
| Device ownership | PC A/B, resource IDs, connection and driver versions | Dispatch + locks |
| Sample and wiring | Sample ID, contacts, Vxx/Vxy mapping, gate, optical path | All stages |
| Electrical convention | Rs calibration, actual I, RMS/peak, f_I, harmonics, phase | Hall analysis |
| Optical convention | Source, excitation spectrum, PD calibration, polarization zero | BVPE analysis |
| Environment | T/B/Vg targets, readbacks and settling criteria | Point barrier |
| Action envelope | Approved drive, optical power, gate leakage and rates | Validation + QC |
| End-state policy | Excitation off; gate and environment follow defined policy | Run closure |

**A setup is both a resource map and an explicit statement of what each measurement means.**

# Two transport lock-ins, with explicit drive and sense paths

The master lock-in supplies AC excitation and Vxx(1f); the follower measures Vxy at a selected harmonic.




**Vxx / Vxy are spatial channels. X / Y are phase components returned by EACH lock-in.**

Circuit-level scenario based on the supplied apparatus; harmonic measurement rationale: [2,3].

# Change acquisition mode, preserve channel meaning

The same two transport instruments support a sequence of complementary harmonic and reference stages.

| Stage | Master: Vxx | Follower: Vxy | Scientific purpose |
|---|---|---|---|
| A: transport baseline | 1f, internal REF | 1f, external f_I | Linear response and pickup baseline |
| B: second-order test | 1f, internal REF | 2f, external f_I | Vxy(2ω) versus actual Iω² |
| C: harmonic control | 2f, internal REF | 2f, external f_I | Longitudinal / transverse comparison |
| D: return reference | Chosen reference mode | Same reference settings | Repeatability and drift check |

## Preserve the full readout

X / Y, frequency, harmonic, gain, range and flags

## Keep inference separate

Quadratic response is evidence, not a unique mechanism

# Turn a nonlinear signal into a controlled comparison

Outer coordinates: T × B × Vg. Inner tests: drive amplitude, frequency, harmonic and return references.

**Prepare state**
T / B / Vg readbacks

→

**Dark baseline**
Low drive + 1f

→

**Drive sweep**
Actual Iω + 2f

→

**Return controls**
Repeat / f / harmonics

→

**Evidence review**
Qualify + compare

## Configured measurements

Current bracket and return point
Frequency / phase / harmonic controls
Environmental and gate-leakage traces
Repeated reference acquisitions

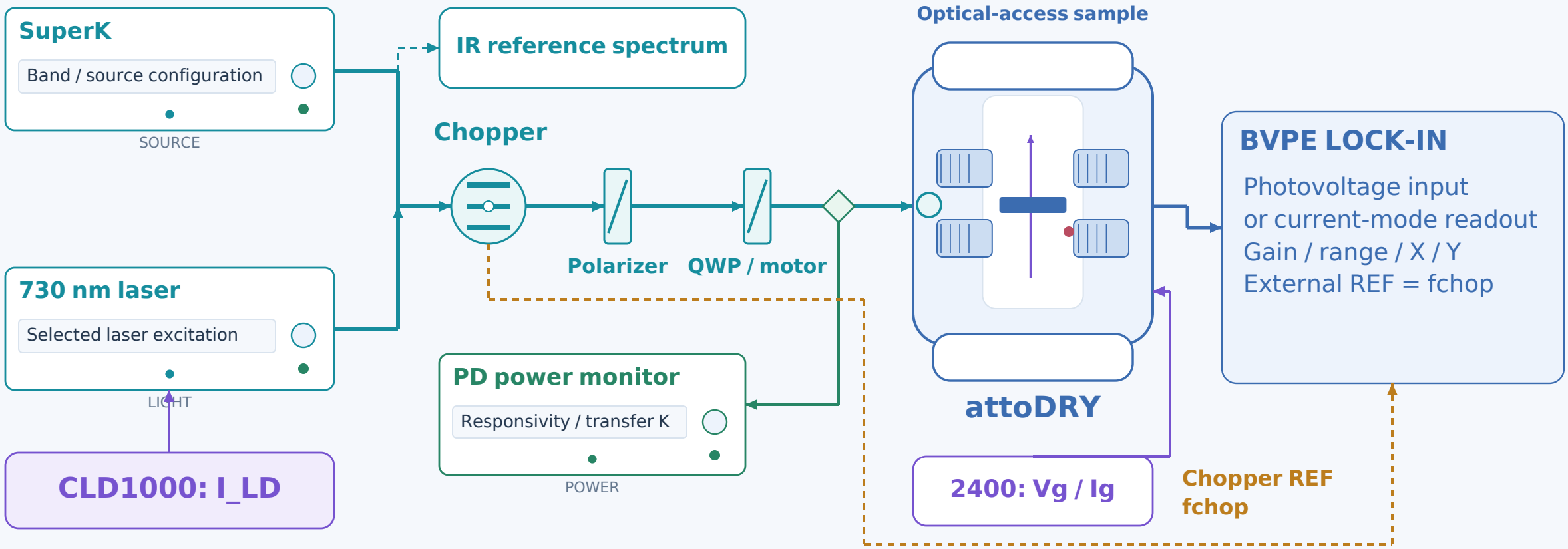
## Evidence for the next decision

Readable and referenced signals?
Consistent response at reference settings?
Drive scaling preserved across controls?
Which comparison resolves ambiguity?

**The agent proposes the next discriminating measurement within the configured experimental envelope.**

# Optical drive, power and referenced readout

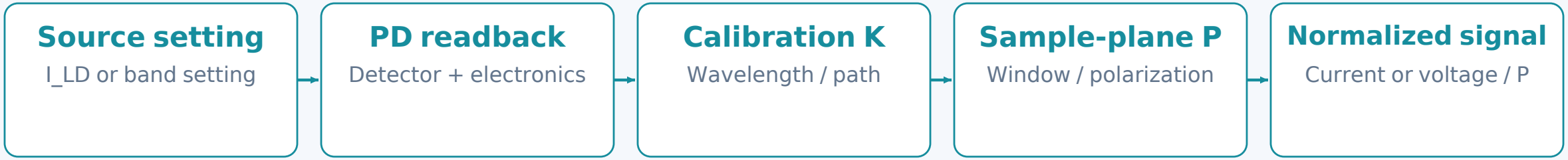
The 730 nm and SuperK paths converge on the same configured polarization and sample environment.




**fchop is the optical-intensity envelope reference; it is not the optical carrier frequency or Hall 2f_I.**

# Describe the optical state at the sample plane

Drive settings, monitor readings and sample-plane power are related through an explicit calibration.




## Power definition

Monitor responsivity
Calibration transfer K
Sample-plane reference
Modulation amplitude convention

## Polarization definition

Polarizer angle zero
QWP angle and retardance
Input polarization state
Motor position readback

## Electrical definition

Voltage or current readout
Transimpedance gain, if used
X / Y phase and lock status
Source-drain bias convention

**For current readout: Iph = Vread / G_TIA. Power normalization must state its calibration and units.**

# Combine spectrum, power, polarization and environment

PC A supplies optical resources; PC B supplies polarization, modulation, sample state and electrical readout.

**Dark / reference**
Background + spectrum

→

**Power calibration**
Source -> PD -> P

→

**Polarization scan**
Linear angle / QWP

→

**State sweep**
T / B / Vg / spectrum

→

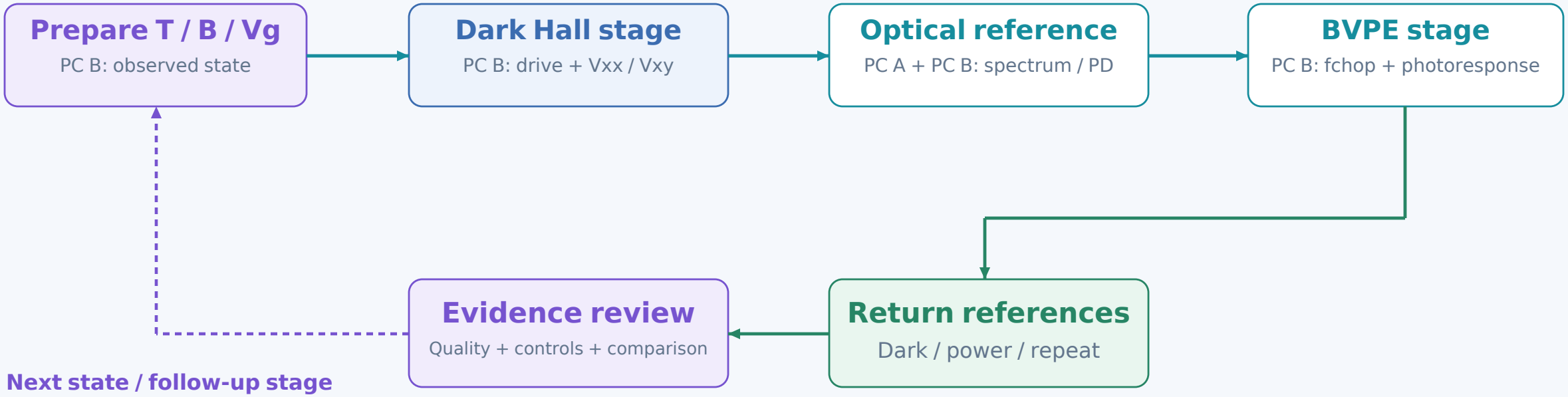
**Return reference**
Dark / P / repeat

| Branch | Configured scan | Associated evidence |
|---|---|---|
| 730 nm excitation | Drive setting × power × polarization | PD, angle, gain and fchop readbacks |
| SuperK excitation | Excitation band / recorded incident spectrum | Reference spectrum and power calibration |
| Environmental sweep | T × B × Vg with selected optical state | Actual state, settling and gate leakage |
| Reference return | Dark / reference power / repeated polarization | Background, stability and repeatability |

**A chopped-light signal is an observation; mechanism assignments require the matching control experiments.**

# A shared coordinate links dark and illuminated stages

campaign_id links the research objective; stage/run IDs preserve branches; point IDs link conditions.




**sample + wiring + observed T/B/Vg + source state + time windows + calibration versions**

**Compare linked conditions and controls before inferring relationships between electrical and optical responses.**

# A measurement is more than a scalar value

Keep requested values, observed state, raw channels, calibration and quality decisions in separate fields.

## Identity and context

campaign / run / stage / point
sample ID and wiring version
plan + setup fingerprints
requested T / B / Vg
observed T / B / Vg / Ig
acquisition start / end times
source + polarization configuration

## Raw measurement channels

Vxx: X, Y, n, f_I
Vxy: X, Y, n, f_I
photoresponse: X, Y, fchop
input mode / gain / range
PD monitor and calibrated power
reference spectrum S(λ)
local file + artifact checksum

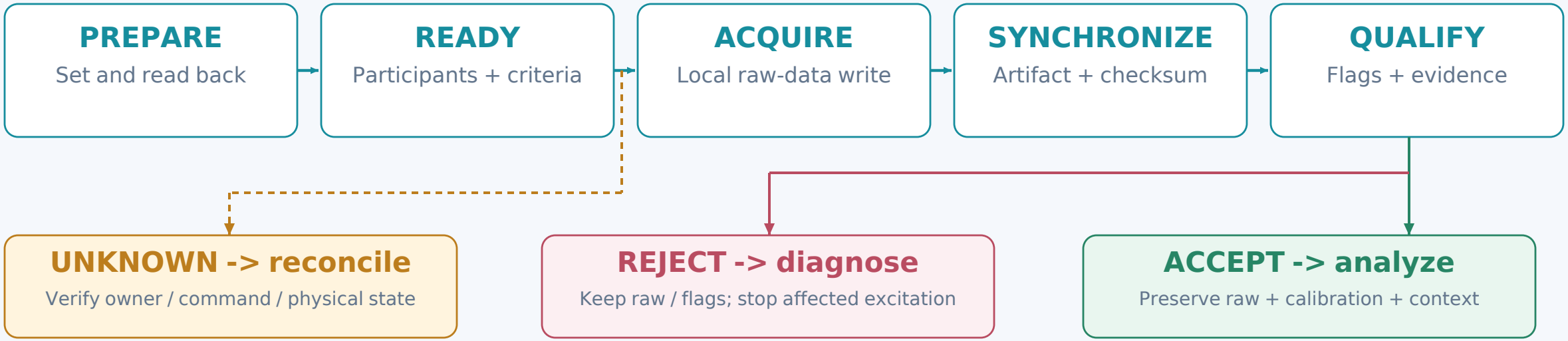
## Qualification and interpretation

reference lock / overload flags
settling / freshness evidence
accepted / rejected / historical
calibration and analysis versions
reference / return comparisons
observations vs. hypotheses
closure state and next decision

**Raw evidence is retained; derived analysis and interpretation remain attributable and revisable.**

# Explicit state transitions govern the next action

A completed command, a synchronized artifact and a scientifically usable reading are distinct outcomes.




## Closure is a recorded state, not just an acknowledgement

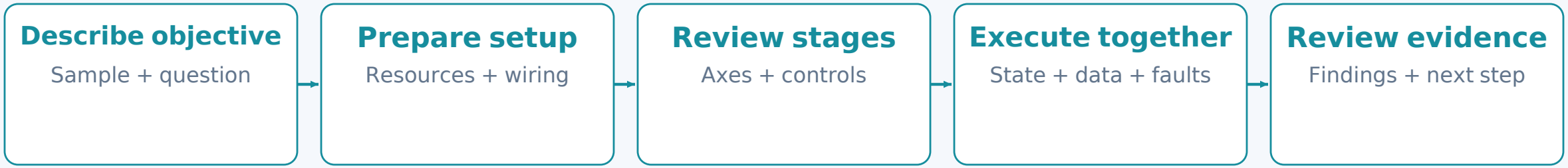
Excitation off and readback verified; gate / T / B follow the selected end-state policy.

**The next proposed stage is validated again and respects operator gates and the configured action envelope.**

# Start from the research objective, not a remote desktop

Device configuration is reusable; each experiment supplies a sample, variables, controls and operating bounds.

**"Map dark nonlinear Hall versus T, B and Vg; then measure 730 nm power and polarization dependence at matching states, with return references."**




PC side: identity / instrument connections / drivers / local data / local stop policy
Experiment side: sample / wiring / allowed ranges / coordinate order / references / closure

**The scientist focuses on design and interpretation; LabOS organizes execution and evidence.**

# Reuse local code. Compose shared capabilities.

An instrument capability is portable at the planning layer even when its vendor environment remains local.

### Existing instrument code

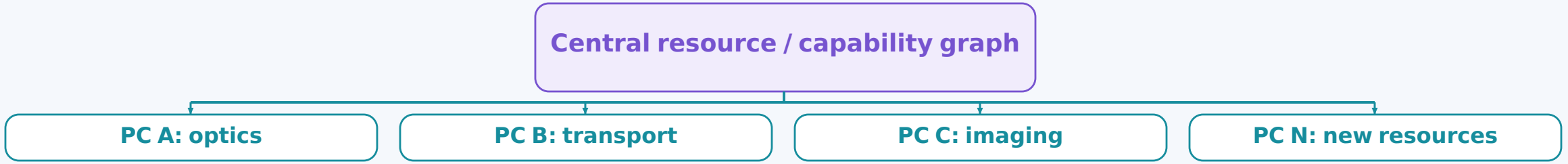
Connect / identify / status
Set / read back / settle
Acquire / write raw data
Stop / verify output state

### Bounded local capability

Typed inputs and units
Defined return data and flags
Timeout / cancellation behavior
Exclusive device session

### Reusable scientific stage

Environment preparation
Hall harmonic acquisition
Optical calibration / BVPE
Reference and diagnostic tests




**Scale by adding owned resources and verified capabilities, not by moving every instrument to one PC.**

LABOS CAPABILITY MAP

# One coordinated experiment. Multiple instruments and PCs.

- **Coordinate** — Optical sources, sample state, transport and modulation
- **Preserve meaning** — References, harmonics, power calibration and timestamps
- **Close the loop** — Qualified observations, controls and a justified next stage

Nonlinear Hall + BVPE: a concrete example of composable, evidence-driven experimentation

# Scientific context and diagram provenance

All architecture, apparatus, circuit, timing and state diagrams are original LabOS scenario drawings.


### [1] Scientific-agent architecture

Rha et al., Autonomous Quantum Measurement with Agentic AI (2026).
https://arxiv.org/abs/2610.38850

### [2] Nonlinear Hall theory

Sodemann and Fu, Physical Review Letters 115, 216806 (2015).
https://doi.org/10.1103/PhysRevLett.115.216806

### [3] Nonlinear Hall measurement

Ma et al., Nature 565, 337-342 (2019).
https://doi.org/10.1038/s41586-018-0807-6

### [4] Bulk photovoltaic response

Osterhoudt et al., Nature Materials 18, 471-475 (2019).
https://doi.org/10.1038/s41563-019-0297-4


**Apparatus and interfaces: the supplied two-PC configuration. Figures explain system capabilities and workflows.**